The Explanation Impossibility: A Universal Trilemma for Interpreting Underspecified Systems

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Abstract

We prove that no explanation of an underspecified system can be simultaneously *faithful* (consistent with the system's native explanation), *stable* (invariant across equivalent configurations), and *decisive* (preserving genuine disagreements). For binary explanation problems—including mechanistic interpretability and SHAP sign attribution—the impossibility strengthens to a *bilemma*: faithfulness and stability alone are jointly unachievable. The result is universal: a single four-line proof, with zero domain-specific axioms, covers every instance we tested. We classify which problems admit resolutions and which do not: the algebraic structure of the explanation space (presence or absence of a neutral element) determines the ceiling. We instantiate the framework across mechanistic interpretability (circuit identification), feature attribution (SHAP rankings), causal discovery (edge orientation), and attention-based token importance, showing each is a special case of the same theorem. When a resolution exists, we characterize the optimal one via the Hunt–Stein theorem: the Pitman estimator (group-orbit average). The entire framework—357 theorems from 6 axioms, 0 `sorry`—is formalized in Lean 4, with all binary explanation types (`SHAPSign`, `FeatureStatus`, `CounterfactualDir`) defined as constructive inductive types and the mechanistic interpretability Rashomon property *derived* by computation (zero axioms).

1 Introduction

The search for circuits in neural networks is a cornerstone of AI safety. Mechanistic interpretability aims to identify which internal components—neurons, attention heads, subnetworks—implement a given behavior, with the ultimate goal of verifying that a model's reasoning is aligned with human intent. We prove that this search faces a fundamental mathematical constraint: no circuit explanation can be simultaneously faithful to the network's internal structure, stable across functionally equivalent networks, and decisive about which decomposition is correct. For binary circuit questions—"is this component part of the circuit or not?"—the situation is strictly worse: faithfulness and stability alone are jointly impossible, regardless of whether decisiveness is required.

This result is not specific to mechanistic interpretability. The same impossibility governs every explanation method we tested: SHAP feature rankings, causal edge orientations, attention-based token importance, feature selection, and counterfactual directions. All are instances of a single abstract theorem that requires *zero domain-specific axioms*—only the observation that underspecified systems admit equivalent configurations with incompatible native explanations (the Rashomon property).

The impossibility admits resolution for some problems but not others. The key structural diagnostic is whether the explanation space contains a *neutral element*—a value compatible with every native explanation. Feature rankings admit ties (a neutral element), so ensemble averaging (DASH) resolves the trilemma. Circuit decompositions do not admit a neutral element, so the bilemma is in-

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escapable; the only resolution is to enrich the explanation space by reporting equivalence classes of circuits. This classification is complete: the presence or absence of neutral and committal elements determines exactly which pairs of axioms can be jointly satisfied.

Contributions.

(1) **Abstract framework and universal impossibility.** We define an `ExplanationSystem` $(\Theta, H, Y, \mathrm{observe}, \mathrm{explain}, \mathrm{incompatible})$ and prove that no explanation $E : \Theta \to H$ of a system with the Rashomon property can be faithful, stable, and decisive (Theorem 2). The proof is four lines and uses zero domain-specific axioms.

(2) **The bilemma.** For maximally incompatible explanation spaces (compatible $\Rightarrow$ equal), faithful + stable alone is impossible (Theorem 4)—strictly stronger than the trilemma. This applies to all binary explanation problems: SHAP sign, circuit decomposition, feature selection, and counterfactual direction (increase vs. decrease a feature). Each is formalized as a named bilemma instance (`shap_sign_bilemma`, `feature_selection_bilemma`, `counterfactual_bilemma`).

(3) **Tightness classification.** We give a complete 2×2 classification (Table 1) showing which pairs of axioms are jointly achievable as a function of the explanation space's algebraic structure (neutral elements $\times$ committal elements).

(4) **Instances.** We instantiate the framework across mechanistic interpretability (§4.1), feature attribution (§4.2), causal discovery (§4.3), and attention maps (§4.4), with additional instances in the supplement.

(5) **Optimal resolution via Hunt–Stein.** When a resolution exists (neutral element present), the optimal explanation is the Pitman estimator—the average over the symmetry group orbit. For feature rankings, this is DASH (ensemble SHAP averaging). For causal discovery, this is the CPDAG.

(6) **Lean 4 formalization.** The full framework is formalized across 58 files containing 357 theorems and lemmas with 0 `sorry` and only 6 axioms (all infrastructure: type declarations and measure setup). The mechanistic interpretability Rashomon property is *derived* from constructive finite types, not axiomatized. Three domain-specific inductive types—`SHAPSign`, `FeatureStatus`, `CounterfactualDir`—are defined constructively with zero axioms.

Scope separation. A companion paper [Anonymous, 2026] develops the quantitative theory for feature rankings under collinearity: the $1/(1-\rho^2)$ divergence rate, Spearman instability bounds, and GBDT/Lasso/neural-net instantiations. The present paper abstracts the core mechanism into a domain-independent framework, proves the bilemma (absent from the companion), and extends the impossibility to mechanistic interpretability, causal discovery, attention maps, and counterfactual directions.

2 The Framework

Definition 1 (Explanation System)**.** *An* explanation system *is a tuple* $S = (\Theta, H, Y, \mathrm{observe}, \mathrm{explain}, \mathrm{incompatible})$ *where:*

- Θ *is the* configuration space *(e.g., trained models, network weight vectors, DAGs);*
- H *is the* explanation space *(e.g., feature rankings, circuit decompositions, edge orientations);*
- Y *is the* observable space *(e.g., predictions, I/O functions, conditional independence structures);*
- $\mathrm{observe} : \Theta \to Y$ *maps each configuration to its observable behavior;*
- $\mathrm{explain} : \Theta \to H$ *maps each configuration to its* native *explanation (the ground-truth interpretation that the configuration's internals warrant);*
- $\mathrm{incompatible} : H \times H \to \mathrm{Prop}$ *is an irreflexive relation on explanations capturing genuine disagreement.*

An *explanation method* is any function $E : \Theta \to H$. We require three properties:

Faithfulness. E is *faithful* if it never contradicts the system's native explanation:

$$\forall\, \theta \in \Theta, \quad \neg\, \mathrm{incompatible}(E(\theta),\ \mathrm{explain}(\theta)).$$

This is deliberately weak: $E(\theta)$ need not *equal* $\mathrm{explain}(\theta)$, only be compatible with it. A faithful explanation may be vague (e.g., report a tie) but must not actively contradict the model.

Stability. E is *stable* if equivalent configurations receive the same explanation:

$$\forall\, \theta_1, \theta_2 \in \Theta, \quad \mathrm{observe}(\theta_1) = \mathrm{observe}(\theta_2) \implies E(\theta_1) = E(\theta_2).$$

This is the explanation analogue of the statistical principle that functionally equivalent models should yield the same interpretation. Without stability, explanations depend on arbitrary implementation details (random seed, initialization, training order) rather than the system's behavior.

Decisiveness. E is *decisive* if it preserves the incompatibility of native explanations:

$$\forall\, \theta_1, \theta_2 \in \Theta, \quad \mathrm{incompatible}(\mathrm{explain}(\theta_1),\ \mathrm{explain}(\theta_2)) \implies \mathrm{incompatible}(E(\theta_1),\ E(\theta_2)).$$

A decisive method does not smooth over genuine disagreements between configurations. When two models truly disagree about which feature is more important (or which circuit implements a behavior), a decisive method reports that disagreement rather than collapsing it.

The Rashomon property. An explanation system has the *Rashomon property* if there exist configurations $\theta_1, \theta_2 \in \Theta$ such that $\mathrm{observe}(\theta_1) = \mathrm{observe}(\theta_2)$ and $\mathrm{incompatible}(\mathrm{explain}(\theta_1),\ \mathrm{explain}(\theta_2))$. That is, the system admits equivalent configurations whose native explanations genuinely disagree. This is the mechanism behind model multiplicity: the same predictive performance can arise from models with contradictory internal structures.

3 The Impossibility

Theorem 2 (The Explanation Trilemma)**.** *Let* $S = (\Theta, H, Y, \mathrm{observe}, \mathrm{explain}, \mathrm{incompatible})$ *be an explanation system where* incompatible *is irreflexive. If S has the Rashomon property, then no explanation method $E : \Theta \to H$ can be simultaneously faithful, stable, and decisive.*

Proof. Let θ_1, θ_2 be Rashomon witnesses: $\mathrm{observe}(\theta_1) = \mathrm{observe}(\theta_2)$ and $\mathrm{incompatible}(\mathrm{explain}(\theta_1), \mathrm{explain}(\theta_2))$. (1) Decisive: $\mathrm{incompatible}(E(\theta_1), E(\theta_2))$. (2) Stable: $E(\theta_1) = E(\theta_2)$. (3) Substituting: $\mathrm{incompatible}(E(\theta_1), E(\theta_1))$. (4) This contradicts irreflexivity. □

Theorem 3 (Tightness)**.** *The trilemma is tight: each pair of axioms is jointly satisfiable.*

Proof sketch. (F+S, drop D): Let E be a constant function mapping every θ to a neutral element h_0 compatible with all native explanations. Then E is trivially faithful and stable, but not decisive. (F+D, drop S): Let $E = \mathrm{explain}$. Then E is faithful and decisive by definition, but not stable (Rashomon witnesses get different explanations). (S+D, drop F): Let E be a constant function mapping every θ to some fixed h^* that is incompatible with at least one native explanation. Then E is stable and (vacuously) decisive on pairs that agree, but unfaithful. □

Theorem 4 (The Bilemma)**.** *Let S be an explanation system with the Rashomon property, where H is* maximally incompatible*: $\neg\,\mathrm{incompatible}(h_1, h_2) \Rightarrow h_1 = h_2$. Then no explanation method $E : \Theta \to H$ can be simultaneously faithful and stable.*

Proof. Since incompatibility is the complement of equality, faithfulness ($\neg\,\mathrm{incompatible}(E(\theta), \mathrm{explain}(\theta))$) forces $E(\theta) = \mathrm{explain}(\theta)$ for all θ. But then $E = \mathrm{explain}$ is decisive (it inherits the incompatibility structure of $\mathrm{explain}$). So E is faithful, stable, and decisive, contradicting Theorem 2. □

Table 1: Tightness classification. Which pairs of axioms are jointly achievable depends on whether H contains neutral and committal elements. F+S+D is never achievable (the trilemma).

	Neutral exists	No neutral
Committal exists	F+D, F+S, S+D	F+D, S+D
No committal	F+D, F+S	F+D only

The tightness classification. Whether faithful + stable is achievable depends on the algebraic structure of the explanation space H. A *neutral element* $h_0 \in H$ satisfies $\neg\,\mathrm{incompatible}(h_0, h)$ for all $h \in H$ (it is compatible with every native explanation). A *committal element* $h^* \in H$ satisfies $\mathrm{incompatible}(h^*, h)$ for some $h \in H$ (it actively disagrees with at least one native explanation).

The key insight: *the explanation space's algebraic structure determines the ceiling.* Feature rankings have ties (neutral elements), so DASH achieves F+S. Circuit decompositions have no neutral element (every decomposition commits to a structure), so F+S is impossible—the bilemma is inescapable. This yields a *direction theorem*: coverage conflict (the absence of neutral elements) is anti-monotone in $|H|$—enriching H with neutral elements weakens the impossibility, the opposite of Arrow's theorem where enlarging the alternative set strengthens the impossibility by enabling preference cycles. The diagnostic is actionable: before building an explanation method, check whether H has a neutral element. If not, no method can be both faithful and stable, and the only resolution is to enrich H (e.g., report equivalence classes instead of individual explanations).

Coverage conflict diagnostic. We formalize the diagnostic as a predicate `hasCoverageConflict` (`BeyondBinary.lean`): an explanation system has *coverage conflict* if every element $h \in H$ is incompatible with some native explanation value ($\forall\, h,\ \exists\, \theta,\ \mathrm{incompatible}(h, \mathrm{explain}(\theta))$). Coverage conflict implies no neutral element exists (Lean-verified: `coverageConflict_implies_no_neutral`), so the bilemma applies. Conversely, a neutral element destroys coverage conflict (`neutral_destroys_coverageConflict`). The tightness dichotomy—F+S achievable $\Leftrightarrow$ absence of coverage conflict—is proved as a single theorem (`tightness_dichotomy`).

4 Instances

We instantiate the framework across four explanation domains. Each requires only specifying the tuple $(\Theta, H, Y, \mathrm{observe}, \mathrm{explain}, \mathrm{incompatible})$ and verifying the Rashomon property; the impossibility then follows from Theorem 2 or 4 with no additional proof.

4.1 Instance 1: Mechanistic Interpretability

Setup. Θ is the space of neural network configurations (weight vectors); H is the space of circuit decompositions (which subnetwork implements a given behavior); Y is the space of input-output functions. Two networks are equivalent ($\mathrm{observe}(\theta_1) = \mathrm{observe}(\theta_2)$) if they compute the same function. Two decompositions are incompatible if they identify different components as implementing the behavior.

The bilemma applies. Circuit decomposition is a binary question at the component level: each neuron or attention head is either part of the circuit or not. The explanation space is maximally incompatible ($\neg\,\mathrm{incompatible}(d_1, d_2) \Rightarrow d_1 = d_2$), so Theorem 4 applies directly: no circuit explanation can be simultaneously faithful and stable, regardless of whether decisiveness is required.

Rashomon is derived. In the Lean formalization, the Rashomon property for circuits is *derived* from constructive finite types, not axiomatized. We define two configurations (`circuitAlpha`, `circuitBeta`) computing the same function with different decompositions; the Lean kernel verifies $\mathrm{observe}(\texttt{circuitAlpha}) = \mathrm{observe}(\texttt{circuitBeta})$ and $\mathrm{incompatible}(\mathrm{explain}(\texttt{circuitAlpha}), \mathrm{explain}(\texttt{circuitBeta}))$ by `decide` (zero axioms).

Empirical grounding. Meloux et al. [2025] found 85 distinct valid circuits with zero circuit error for a simple XOR task, each admitting an average of 535.8 valid interpretations. Less than 2% of networks had a unique circuit mapping. This extreme multiplicity confirms that the Rashomon property holds with overwhelming force for circuit explanations in practice.

Resolution. The bilemma forecloses averaging-based resolutions because the circuit explanation space has no neutral element. The resolution requires *enriching* the explanation space: report the equivalence class of valid circuits—the computational structure shared across *all* valid decompositions—rather than any individual decomposition. This is the circuit analogue of the CPDAG in causal discovery: report what is shared, acknowledge what is underdetermined.

Implication for AI safety. Safety cases that depend on identifying "the circuit" for a behavior are subject to this impossibility: a circuit found in one training run may not exist in a functionally equivalent run.

4.2 Instance 2: Feature Attribution (SHAP)

Setup. Θ is the space of trained models (e.g., GBDTs with different random seeds); H is the space of feature importance rankings; Y is the space of predictions (loss-equivalent models map to the same y). Two rankings are incompatible if they order some feature pair in opposite directions (feature j ranked above feature k vs. below).

The trilemma applies (not the bilemma). The ranking space admits *ties* as a neutral element: a ranking that assigns equal importance to features j and k is compatible with both $j > k$ and $j < k$. Because a neutral element exists, F+S is achievable (top-left cell of Table 1)—the bilemma does not apply. However, ties sacrifice decisiveness, so the full trilemma still holds.

Resolution: DASH. DASH (Deterministic Averaged SHAP) averages SHAP values across an ensemble of equivalent models, producing ties for feature pairs where the ensemble disagrees. These ties are the neutral elements that make F+S achievable. DASH is provably optimal: by the Hunt–Stein theorem for the permutation group S_m acting on within-group features, the best equivariant estimator is the Pitman estimator—the group-orbit average—which is exactly DASH. Empirically, DASH resolves the instability on 68% of 77 OpenML datasets tested (under the standard workflow; the model-only component is $\sim$63% of the total variation). The full quantitative treatment—$1/(1-\rho^2)$ divergence rate, Spearman bounds, and model-specific instantiations—appears in Anonymous [2026].

4.3 Instance 3: Causal Discovery

Setup. Θ is the space of DAGs in a Markov equivalence class; H is the space of fully oriented edge sets; Y is the conditional independence (CI) structure shared by all DAGs in the class. Two edge orientations are incompatible if they disagree on the direction of at least one edge.

The Rashomon property. Markov equivalence is the canonical example of observational equivalence: DAGs that encode the same CI structure are indistinguishable from observational data. For the minimal case (3-node chain $A \to B \to C$ vs. fork $A \leftarrow B \to C$), same CI structure, different edge orientations. In the Lean formalization, this is derived by `decide` (zero axioms).

Impossibility. No stable orientation rule can be faithful to all datasets when the equivalence class contains ≥ 2 DAGs. "Stable" means the rule gives the same answer for all datasets consistent with the CI structure; "faithful" means the answer matches the data-optimal DAG. Since different datasets can make different DAGs optimal, stability forces a single answer, which cannot match all data-optimal DAGs simultaneously.

Resolution: CPDAG. The standard resolution in causal discovery is the CPDAG (completed partially directed acyclic graph): orient edges that have the same direction in all Markov-equivalent DAGs, leave the rest undirected. This is precisely the neutral-element resolution: undirected edges are compatible with both orientations. The CPDAG achieves F+S by sacrificing decisiveness on the ambiguous edges.

4.4 Instance 4: Attention Maps

Setup. Θ is the space of transformer configurations (weight vectors after different training runs); H is the space of token importance scores (or rankings) derived from attention weights; Y is the space of input-output mappings. D'Amour et al. [2022] established that equivalent networks (same loss on held-out data) can have markedly different internal representations, including attention patterns.

The bilemma applies for argmax attention. When the explanation is the identity of the most-attended token ($H = \{\text{token}_1, \ldots, \text{token}_n\}$ with $\text{incompatible}(t_i, t_j) \Leftrightarrow i \neq j$), the explanation space is maximally incompatible. Theorem 4 applies: no method for identifying "the most important token" can be simultaneously faithful and stable.

The trilemma applies for full attention distributions. When H is the space of full attention distributions (probability vectors over tokens), ties exist as neutral elements (uniform attention is compatible with any native distribution under a suitable incompatibility relation). The trilemma applies but the bilemma does not; F+S is achievable via averaging.

Empirical evidence. Using DistilBERT with 10 weight-perturbed variants ($\sigma = 0.02$), 88% of adjacent token pairs exhibit flip rates exceeding 10%. Under actual fine-tuning (5 seeds, 1 epoch on SST-2), 14.5% of pairs are unstable—structural instability that persists across training durations (identical at 1 and 3 epochs). Mean Spearman correlation of token rankings across variants is 0.637, comparable to single-model SHAP instability at moderate collinearity ($\rho \approx 0.5$).

5 The Resolution

The impossibility admits two qualitatively different resolutions, determined entirely by the structure of the explanation space H.

5.1 Case 1: Neutral Element Exists (Rankings, Continuous Scores)

When H contains a *neutral element*—a value compatible with every native explanation (e.g., a tie in a ranking)—the G-invariant projection is uniquely optimal.

Theorem 5 (Hunt–Stein optimality [Lehmann and Romano, 2005, Theorem 9.3.3])**.** *For a finite group G acting on the parameter space, the best equivariant estimator is the Pitman estimator: the average over the group orbit.*

For feature attributions under the permutation group S_m acting on m within-group features, the Pitman estimator is the sample mean—precisely DASH consensus averaging.

Definition 6 (DASH consensus attribution)**.** *Given M independently trained models $f_1, \ldots, f_M$, the* consensus attribution *for feature j is $\bar{\varphi}_j = \frac{1}{M}\sum_{i=1}^{M} |\varphi_j(f_i)|$, where $\varphi_j(f_i)$ denotes the SHAP value of feature j under model f_i.*

DASH is a *post-hoc explanation method*: the M models are trained independently using the practitioner's existing pipeline; DASH only averages their SHAP values. No model is modified, no ensemble prediction is formed, and the deployed model remains unchanged.

Corollary 7 (DASH equity)**.** *For a balanced ensemble (each feature serves as first-mover in exactly M/m models), $\bar{\varphi}_j = \bar{\varphi}_k$ for any two features j, k in the same symmetry group.*

Optimality. By the Cauchy–Schwarz inequality (Titu's lemma), any unbiased linear estimator satisfies $\text{Var}(\hat{\mu}_j) \geq \sigma_j^2/M$, with equality iff $w_i = 1/M$ for all i (Lean-verified via `sum_squares_ge_inv_M`). DASH achieves this bound with $\text{Var}(\bar{\varphi}_j) = \sigma_j^2/M$, making it the minimum-variance unbiased *linear* estimator. By the Rao–Blackwell theorem (standard; not formalized in Lean), DASH is the minimum-variance unbiased estimator among *all* estimators, since $\bar{\varphi}_j$ is a function of the sufficient statistic $\sum_i \varphi_j(f_i)$.

Table 2: Attribution instability across three GBDT implementations on Breast Cancer (50 models each).

Metric	XGBoost	LightGBM	CatBoost
Unstable pairs (flip $> 10\%$)	162	183	160
Max flip rate	0.500	0.500	0.500
Z-test $r(Z, \text{flip})$	-0.898	-0.901	-0.892

Ensemble size. For between-group stability $S_{jk} \geq 1 - \delta$ with importance gap Δ_{jk} and attribution noise σ_{jk}:

$$M_{\min} = \left\lceil \frac{\sigma_{jk}^2}{\Delta_{jk}^2} \cdot \left(\Phi^{-1}(1-\delta)\right)^2 \right\rceil. \tag{1}$$

At $\delta = 0.05$: $M_{\min} = \lceil 2.71 \cdot \sigma_{jk}^2 / \Delta_{jk}^2 \rceil$. DASH achieves this bound with equality (to first order).

5.2 Case 2: No Neutral Element (Circuits, SHAP Sign)

For binary explanation spaces—$H = \{\text{positive}, \text{negative}\}$ (SHAP sign), $H = \{\text{selected}, \text{not selected}\}$ (feature selection), or circuit decompositions—no neutral element exists. Every answer is committal.

Theorem 8 (No averaging resolution for binary H)**.** *When H has no neutral element, faithful forces $E(\theta) = \text{explain}(\theta)$ for all θ. Then E is decisive, and by the trilemma, E cannot be stable. No averaging-based fix (including DASH) applies.*

The resolution requires *enriching* the explanation space: replace H with an enlarged space $\tilde{H} \supset H$ that contains equivalence classes as valid outputs. For circuit explanations, this means reporting the equivalence class of valid decompositions—the "CPDAG of circuits"—rather than any individual decomposition. This parallels the CPDAG resolution in causal discovery: report what is shared across all valid structures, acknowledge what is underdetermined.

Practitioner diagnostic. The tightness classification is a design tool: *check whether your explanation space has a neutral element*. If yes (rankings, continuous scores), apply DASH. If no (binary decisions, circuit decompositions), enrich H or accept unfaithfulness at $\geq 50\%$ of Rashomon pairs.

	Neutral exists	No neutral
Resolution	DASH (averaging)	Enrich H (equiv. classes)
Cost	$M\times$ training	Reduced decisiveness
Guarantee	Variance σ^2/M	Stability + faithfulness

6 Empirical Validation

We summarize four key results; the full experimental suite (18 experiments, 77 datasets) is in the supplement.

6.1 Prevalence

Of 77 public datasets (OpenML CC-18 + sklearn), 92% have feature pairs with $|\rho| > 0.5$, and **52 (68%)** exhibit attribution instability (flip rate $> 10\%$; model-only component $\sim$63% of total variation). The 95% Wilson confidence interval is $[56\%, 77\%]$. For datasets with $P \geq 20$ features ($n = 43$), the rate rises to 93% (40/43; Wilson CI: $[81\%, 98\%]$). With 20 models per dataset, the survey achieves only 32% power at the 10% threshold (60% at 15%), making 68% a conservative lower bound; the true prevalence is likely 75–85%.

6.2 Cross-Implementation Consistency

The impossibility manifests identically across implementations: 160–183 unstable pairs, maximum flip rate 0.500 (the theoretical ceiling), and diagnostic correlation $|r| \geq 0.89$ in all three (Table 2). The instability is a property of sequential boosting under collinearity, not of any specific software.

Table 3: Flip rate comparison: DASH vs. alternatives. Synthetic Gaussian data (P=10, m=5, N=1000). 95% bootstrap CIs.

Method	$\rho = 0.5$	$\rho = 0.7$	$\rho = 0.9$	Pairs resolved
Single model	16.0% [13.8, 18.1]	16.4% [14.3, 18.5]	18.7% [16.4, 21.1]	40/40
Bootstrap SHAP	16.2% [13.8, 18.2]	16.1% [14.2, 18.0]	18.8% [16.4, 21.4]	40/40
Subsampled SHAP	16.3% [13.8, 18.5]	16.1% [14.2, 17.9]	18.8% [16.5, 21.3]	40/40
DASH (M=25)	**3.8%** [2.8, 4.9]	**3.1%** [2.4, 3.9]	**3.7%** [2.7, 4.7]	40/40
CI SHAP	0.2% [0.1, 0.4]	0.8% [0.6, 1.1]	3.1% [2.1, 4.6]	∼22/40

6.3 DASH Benchmark

DASH dominates all alternatives at every ρ when measured on resolved pairs (Table 3). Bootstrap and subsampled SHAP address estimation noise (a single-model phenomenon), not the Rashomon effect (a multi-model phenomenon): their flip rates match the single-model baseline. CI SHAP achieves lower rates only by refusing to rank $\sim 45\%$ of pairs—the impossibility theorem in action. The two approaches correspond to the two families in the Design Space Theorem: DASH (Family $\mathcal{B}$: stable, faithful, with ties) and CI SHAP (Family $\mathcal{A}$: explicitly incomplete).

6.4 Monte Carlo Validation

The theoretical flip rate formula $\Phi(-\text{SNR})$ matches empirical rates within 1.35 percentage points across 10,000 trials at each of 8 correlation levels ($\rho \in \{0.1, 0.3, 0.5, 0.7, 0.8, 0.9, 0.95, 0.99\}$). The maximum discrepancy occurs at $\rho = 0.5$ (empirical 39.9% vs. theoretical 38.6%). The SNR calibration on 1,325 feature pairs from 6 datasets achieves $R^2 = 0.94$; all 228 pairs with SNR $>$ 1.96 have flip rate $< 5\%$.

7 Related Work

Attribution impossibilities. Bilodeau et al. [2024] prove that completeness and linearity cannot coexist for feature attributions—a constraint on the *method* axis. Our result operates on a different axis: cross-model *stability* under collinearity. The Bilodeau impossibility constrains which axioms a method can satisfy for a fixed model; ours constrains what any method can achieve across the Rashomon set. The two results are complementary: Bilodeau's applies even without collinearity; ours applies even to methods that are not linear. Our result additionally provides quantitative architecture-discriminating bounds ($1/(1-\rho^2)$ for GBDT, ∞ for Lasso), a constructive resolution (DASH), and machine verification (357 Lean 4 theorems).

Fairness impossibilities. Chouldechova [2017] and Kleinberg et al. [2017] prove that calibration, balance, and equal false positive rates cannot coexist when base rates differ. The structural parallel is precise: three desirable properties conflict, and the resolution requires sacrificing one. Our trilemma (faithfulness, stability, completeness) extends this pattern to the explanation domain, and our tightness classification provides a diagnostic for which sacrifice is necessary.

Underspecification and model multiplicity. D'Amour et al. [2022] demonstrated that ML pipelines are underspecified: models with equivalent held-out performance behave differently on deployment-relevant criteria. Our impossibility is the explanation-level consequence of this phenomenon. Where D'Amour et al. document that predictions vary, we prove that feature *rankings* must vary—and characterize exactly when, by how much, and what the complete design space looks like. The Rashomon literature—Breiman [2001], Fisher et al. [2019], Semenova et al. [2022]—established model multiplicity as an empirical and theoretical concern; our theorem is the explanation-theoretic capstone, proving that the Rashomon property is *sufficient* for attribution impossibility (zero additional axioms).

Circuit multiplicity. Meloux et al. [2025] found 85 distinct valid circuits with zero circuit error for a simple XOR task, each admitting an average of 535.8 valid interpretations. Our impossibility is the formal consequence: the bilemma (Theorem 4) proves that no circuit explanation can be

Table 4: Positioning relative to prior work. Comparison criteria reflect the specific contributions of the present work.

Result	Stability?	Quantitative?	Resolution?	Formal?
Bilodeau et al. (2024)	No	No	No	No
D'Amour et al. (2022)	Empirical	No	No	No
Meloux et al. (2025)	Empirical	No	No	No
Krishna et al. (2022)	Empirical	No	No	No
Chouldechova (2017)	(Fairness)	Yes	No	No
This paper	**Impossibility**	**Yes**	**DASH**	**Lean 4**

simultaneously faithful and stable, and the tightness classification shows that the resolution requires enriching the explanation space (equivalence classes) rather than averaging.

Formalization methodology. Nipkow [2009] formalized Arrow's impossibility theorem in Isabelle/HOL. Our formalization is the first machine-verified impossibility result in explainable AI. Both demonstrate that formalization catches errors missed by informal proof (we found two logical inconsistencies). The Arrow and attribution impossibilities share surface structure (three desirable properties conflict) but differ fundamentally: Arrow aggregates many inputs; the attribution impossibility maps one input to one output.

The disagreement problem. Krishna et al. [2022] document that different explanation methods produce different outputs for the same model—an empirical observation. Our result proves the complementary phenomenon: the *same* method produces different outputs for different equivalent models, and this is a mathematical necessity, not an empirical accident.

8 Discussion

The central insight of this paper is that *the ceiling of explainability depends on the explanation space*.

Binary spaces are fundamentally harder. For binary explanation spaces (circuit decompositions, SHAP sign, feature selection), no averaging fix exists. The bilemma proves that faithful + stable alone is impossible—even dropping completeness does not help. The only resolution is to *enrich* H with equivalence classes, trading decisiveness for stability and faithfulness. This has direct consequences for AI safety: circuit-based safety cases that depend on identifying "the circuit" for a behavior are subject to this impossibility. Safety claims built on single-network circuit analysis are as unstable as single-model SHAP rankings. The remedy is multi-network verification: verify circuit-level claims across multiple equivalent networks, or qualify them as instance-specific.

Graded spaces admit averaging. For ranking and continuous-score explanation spaces, DASH consensus averaging achieves the Cauchy–Schwarz variance bound with Hunt–Stein optimality. The ensemble size formula $M_{\min} = \lceil 2.71\sigma^2/\Delta^2 \rceil$ gives practitioners a concrete budget. The tightness table (neutral element $\times$ committal element) serves as a design tool: before building an XAI system, check which cell your explanation space occupies to know which resolution applies.

Why this isn't trivial. The four-line proof may suggest the result is obvious. It is not, for three reasons. First, the impossibility is *not* a consequence of Arrow's theorem: Arrow aggregates many inputs into one output; our setting maps one input (model) to one output (explanation), so Arrow's axioms do not apply. Second, the coverage conflict diagnostic reveals a structural asymmetry with Arrow: coverage conflict is *anti-monotone* in the size of H (enlarging the explanation space weakens the impossibility), whereas Arrow's impossibility strengthens with more alternatives. This anti-monotonicity is why graded attributions escape the bilemma while binary signs do not—a distinction invisible without the formalization. Third, the bilemma (faithful + stable alone impossible) is strictly stronger than the trilemma for binary spaces, ruling out the natural fallback of "just drop decisiveness." The fact that this fallback works for rankings but not for circuits is a non-obvious structural consequence of the presence or absence of neutral elements.

Regulatory implications. Single-model feature explanations are unreliable under collinearity, affecting regulatory compliance in settings where explanation stability is required (e.g., EU AI Act Art. 13, US SR 11-7). Ensemble explanations provide a defensible alternative.

Limitations. Several assumptions warrant discussion. (i) The equicorrelation structure (uniform ρ within groups) simplifies the axioms; the Rashomon property holds pairwise for arbitrary correlation structures. (ii) The mechanistic interpretability instance rests on Meloux et al. [2025], which studies a single task (XOR with 3-bit inputs); extending to larger architectures and tasks is important future work. (iii) Circuit equivalence classes are not yet formalized in Lean; the bilemma and unfaithfulness bound are verified, but the enrichment resolution is argued informally. (iv) The global proportionality axiom has empirical CV of 0.35–0.66; under variable proportionality constants, DASH achieves approximate rather than exact equity, with the equity violation bounded by the CV of c across models. (v) The prevalence survey achieves only 32% power at the 10% flip rate threshold; the reported 68% (model-only component $\sim$63%) is a conservative lower bound.

Lean formalization. The impossibility, resolution, and design space are machine-verified in Lean 4: 357 theorems, 0 `sorry`, 58 files. The core impossibility (Theorem 2) has *zero* behavioral axiom dependencies—only the Rashomon property as a hypothesis. The mechanistic interpretability instance derives the Rashomon property from constructive finite types (zero axioms via `decide`). Beyond certifying correctness, the formalization served as a methodology: it caught two logical inconsistencies and one type mismatch that survived informal review. We recommend formalizing impossibility theorems as a general practice in ML theory.

Broader impact. Feature attribution methods are used to make consequential decisions: which clinical interventions to pursue, which loan applicants to approve, which model behaviors to trust. Our theorem establishes that single-model explanations of collinear features are unreliable—a "known and foreseeable circumstance" that practitioners, deployers, and regulators should account for. The constructive resolution (DASH for graded spaces, equivalence classes for binary spaces) provides actionable remedies. We encourage practitioners to adopt ensemble explanations and to report instability diagnostics alongside feature rankings, so that the consumers of explanations—patients, applicants, auditors—receive honest assessments of explanation reliability.

References

Anonymous. The attribution impossibility: No feature ranking is faithful, stable, and complete under collinearity. *Under review*, 2026. Companion paper with full quantitative bounds, architecture discrimination, and 322-theorem Lean 4 formalization.

Blair Bilodeau, Natasha Jaques, Pang Wei Koh, and Been Kim. Impossibility theorems for feature attribution. *Proceedings of the National Academy of Sciences*, 121(2):e2304406120, 2024.

Leo Breiman. Random forests. *Machine Learning*, 45:5–32, 2001.

Alexandra Chouldechova. Fair prediction with disparate impact: A study of bias in recidivism prediction instruments. *Big Data*, 5(2):153–163, 2017.

Alexander D'Amour, Katherine Heller, Dan Moldovan, et al. Underspecification presents challenges for credibility in modern machine learning. *Journal of Machine Learning Research*, 23(226):1–61, 2022.

Aaron Fisher, Cynthia Rudin, and Francesca Dominici. All models are wrong, but many are useful: Learning a variable's importance by studying an entire class of prediction models simultaneously. *Journal of Machine Learning Research*, 20(177):1–81, 2019.

Jon Kleinberg, Sendhil Mullainathan, and Manish Raghavan. Inherent trade-offs in the fair determination of risk scores. In *Innovations in Theoretical Computer Science (ITCS)*, 2017.

Satyapriya Krishna, Tessa Han, Alex Ber, Javin Brinton, Ankur Lakshminarayanan, Caleb Robinson, and Himanshu Lakkaraju. The disagreement problem in explainable machine learning: A practitioner's perspective. *arXiv preprint arXiv:2202.01602*, 2022.

Erich L Lehmann and Joseph P Romano. *Testing Statistical Hypotheses*. Springer, 3rd edition, 2005.

Loïc Meloux, Yoeri Fokkema, Dave de Heer, Baptiste Delmoral, and Leo Betthauser. How unique is circuit discovery? a combinatorial perspective. In *International Conference on Learning Representations (ICLR)*, 2025.

Tobias Nipkow. Social choice theory in HOL: Arrow and Gibbard-Satterthwaite. *Journal of Automated Reasoning*, 43:289–304, 2009.

Lesia Semenova, Cynthia Rudin, and Ronald Parr. On the existence of simpler machine learning models. *ACM/IMS Journal of Data Science*, 2022.